Chapter 1: Data Warehousing

(Chapter 11) Modern Database Management 6th Edition

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Content

1. Basic Concepts of data warehousing
2. Data warehouse (DW) architectures
3. Some characteristics of DW data
4. The reconciled data layer
5. Data transformation
6. The derived data layer
7. The user interface
8. Summary

1. Basic Concepts: Motivation

- **"Modern organization is drowning in data but starving for information".**
- ***Operational processing* (transaction processing) captures, stores and manipulates data to support daily operations.**
- ***Information processing* is the analysis of data or other forms of information to support decision making.**

- ***Data warehouse* can consolidate and integrate information from many internal and external sources and arrange it in a meaningful format for making business decisions.**

1. Basic Concepts: Definition

- **Data Warehouse**: (W.H. Immon)
  - A subject-oriented, integrated, time-variant, non-updatable collection of data used in support of management decision-making processes
  - ***Subject-oriented:*** e.g. customers, patients, students, products
  - ***Integrated:*** Consistent naming conventions, formats, encoding structures; from multiple data sources
  - ***Time-variant:*** Can study trends and changes
  - ***Nonupdatable:*** Read-only, periodically refreshed
- **Data Warehousing**:
  - The process of constructing and using a DW

1. Basic Concepts: DW—Subject-Oriented

- Organized around major subjects, such as customer, product, sales.
- Focusing on modeling and analysis of data for decision makers, not on daily operations or transaction processing.
- Provide a simple and concise view around particular subject issues by excluding data that are not useful in the decision support process.

1. Basic Concepts: DW - Integrated

- Constructed by integrating multiple, heterogeneous data sources
  - relational databases, flat files, on-line transaction records
- Data cleaning and data integration techniques are applied.
  - Ensure consistency in naming conventions, encoding structures, attribute measures, etc. among different data sources
    - E.g., Hotel price: currency, tax, breakfast covered, etc.
  - When data is moved to the warehouse, it is converted.

1. Basic Concepts: DW -Time Variant

- The time horizon for the DW is significantly longer than that of operational systems.
  - Operational database: current value data.
  - DW data: provide information from a historical perspective (e.g., past 5-10 years)
- Every key structure in the DW
  - Contains an element of time, explicitly or implicitly
  - But the key of operational data may or may not contain “time element”.

1. Basic Concepts: DW-Non Updatable

- A physically separate store of data transformed from the operational environment.

- Operational update of data does not occur in the DW environment.
  - Does not require transaction processing, recovery, and concurrency control mechanisms.
  - Requires only two operations in data accessing:
    - ✓ *initial loading of data* and
    - ✓ *access of data*.

1. Basic Concepts: Need for Data Warehousing

- Integrated, company-wide view of high-quality information (from disparate databases)
- Separation of *operational* and *informational* systems and data (for improved performance)

Table 11-1: comparison of operational and informational systems

Table 11-1 Comparison of Operational and Informational Systems

Characteristic	*Operational Systems*	*Informational Systems*
Primary purpose	Run the business on a current basis	Support managerial decision making
Type of data	Current representation of state of the business	Historical point-in-time (snapshots) and predictions
Primary users	Clerks, salespersons, administrators	Managers, business analysts, customers
Scope of usage	Narrow, planned, and simple updates and queries	Broad, ad hoc, complex queries and analysis
Design goal	Performance throughput, availability	Ease of flexible access and use
Volume	Many, constant updates and queries on one or a few table rows	Periodic batch updates and queries requiring many or all rows

1. Basic Concepts: Need to separate operational and information systems

Three primary factors:

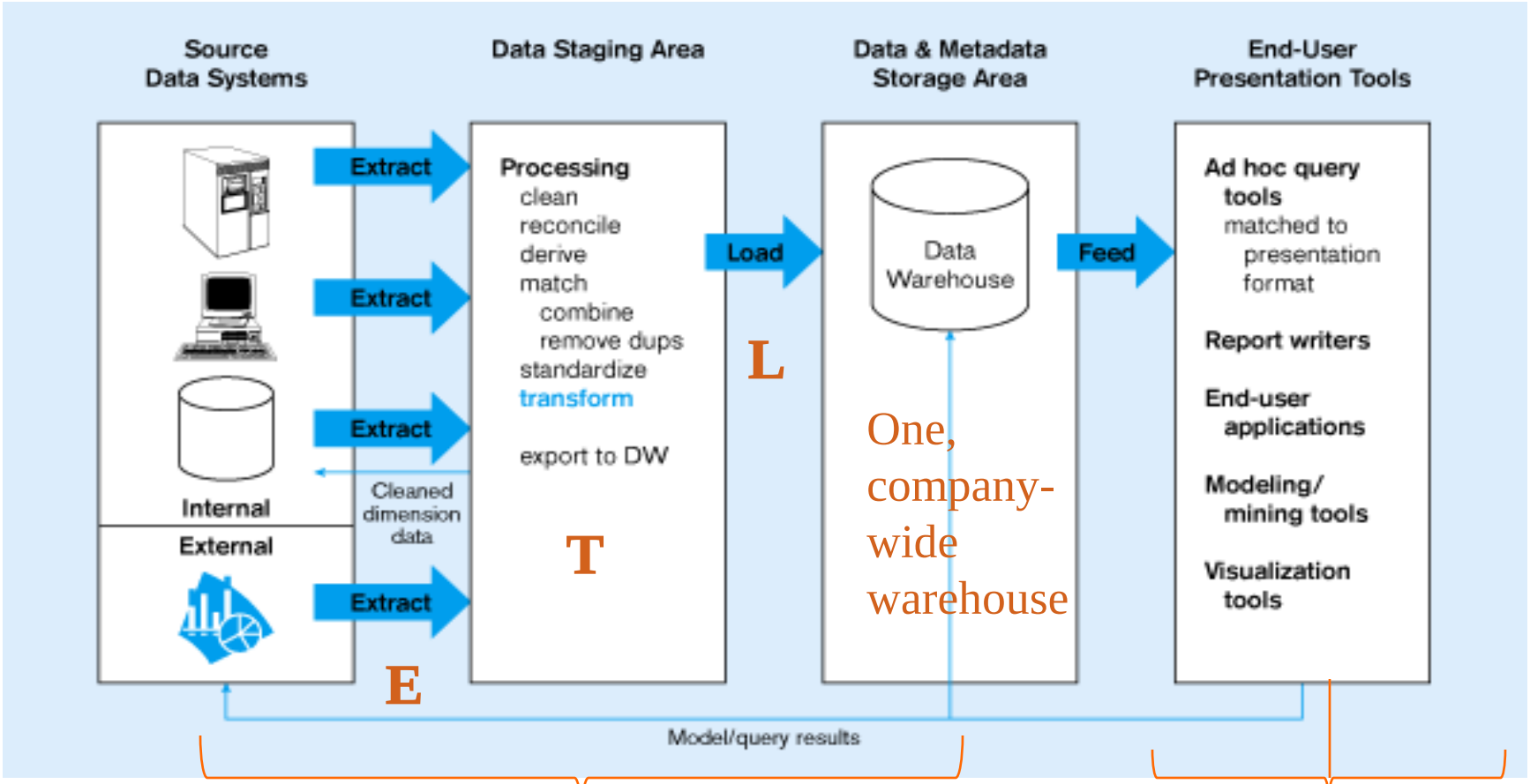
- A DW centralizes data that are scattered throughout disparate operational systems and makes them available for DS
- A well-designed DW adds value to data by improving their quality and consistency
- A separate DW eliminates much of the contention for resources that results in when information applications are mixed with operational processing

2. DW Architectures

1. Generic Two-Level Architecture
2. Independent Data Mart
3. Dependent Data Mart and Operational Data Store
4. Logical Data Mart and @ctive Warehouse
5. Three-Layer architecture

All involve some forms of ***extraction***, ***transformation*** and ***loading*** (**ETL**)

Figure 11-2: Generic two-level architecture



Periodic extraction ➔ data is not completely current in warehouse

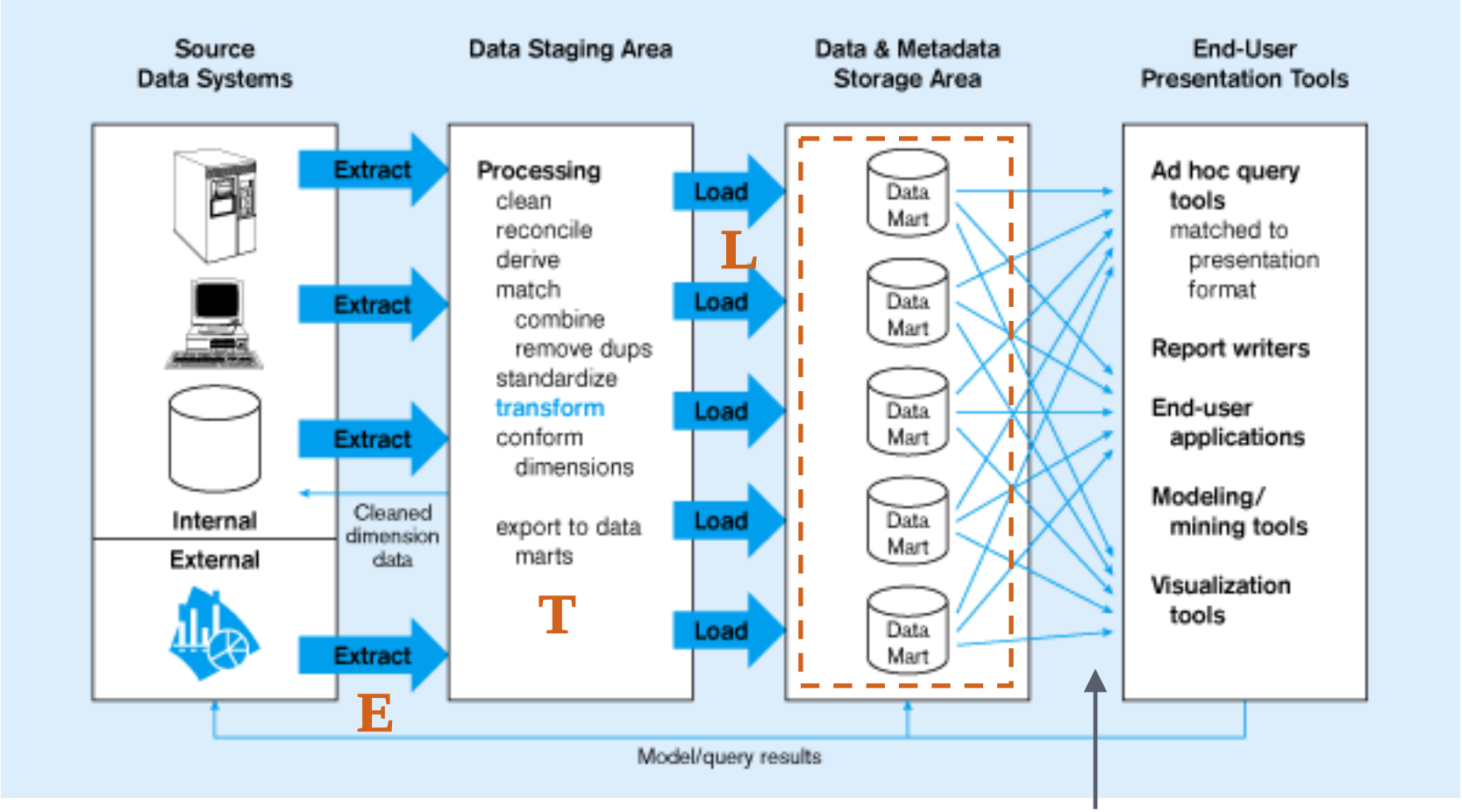
2. DW Architectures: Two-Level Architecture

- **Bottom Tier:** data from different source systems are extracted and loaded to Staging area where all the data is cleaned and changed to standard or common format and transformations are applied to the data and is loaded to DW
- **Top Tier:** the reporting layer, where users execute their reporting queries and use the results in decision making

Figure 11-3: Independent Data Mart

Data marts:

Mini-warehouses, limited in scope



Separate ETL for each ***independent*** data mart

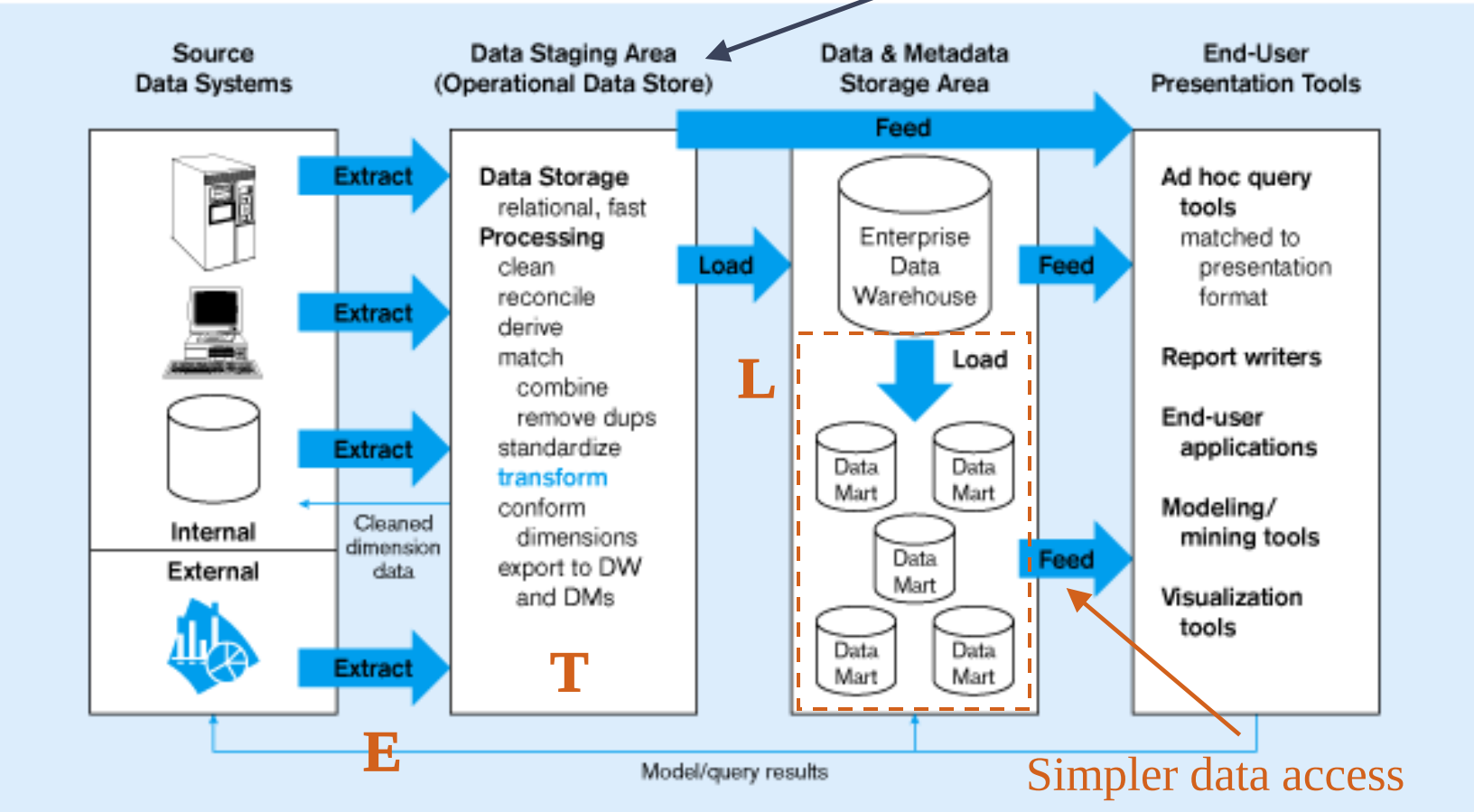
Data access complexity due to ***multiple*** data marts

2. DW Architecture: Independent Data mart

- Independent data mart: a data mart filled with data extracted from the operational environment without benefits of a DW

Figure 11-4:
Dependent data mart with *operational data store*

ODS provides option for obtaining ***current*** data



Single ETL for
enterprise data warehouse (EDW)

Dependent data marts loaded from EDW

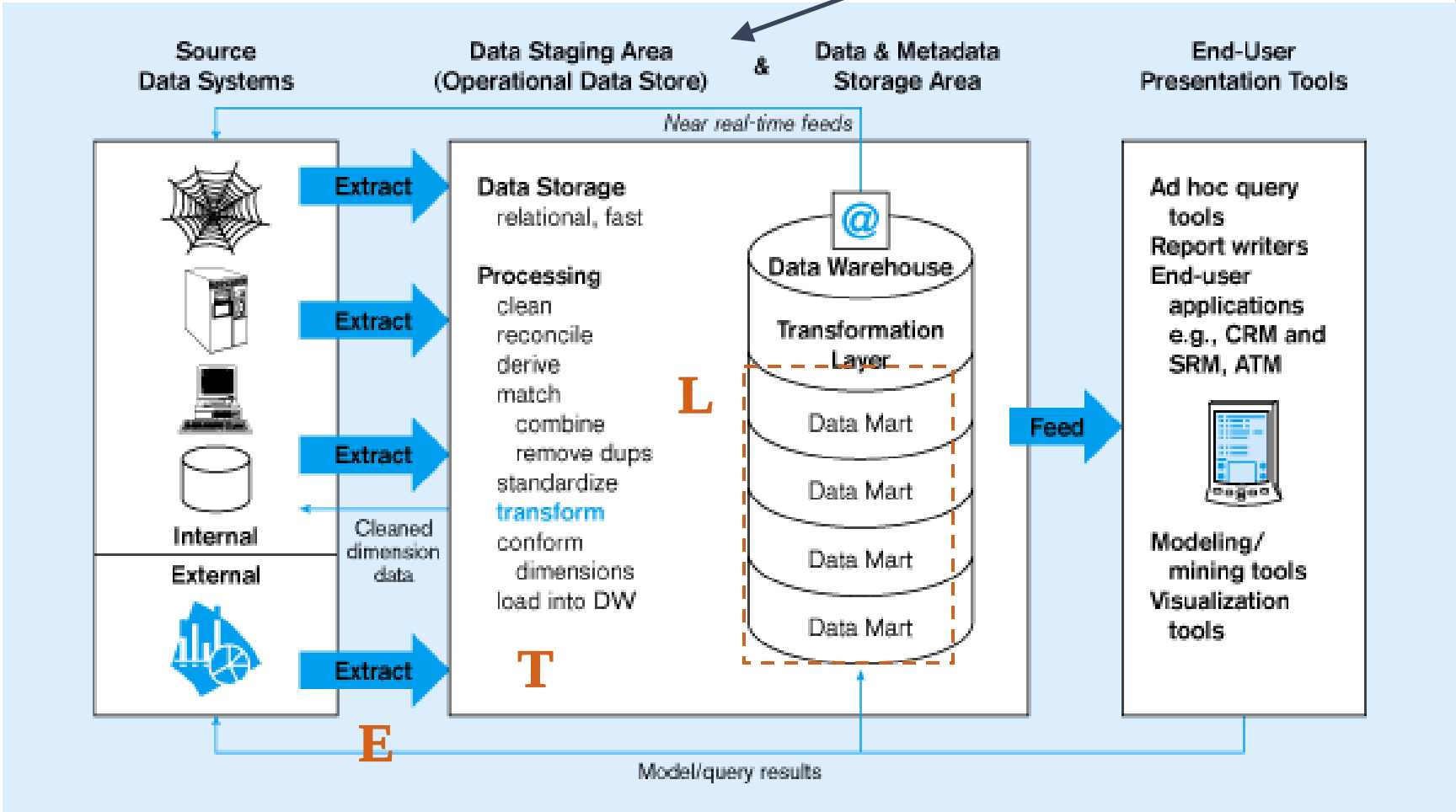
2. DW Architecture: Dependent data mart- Operational data store

- **Dependent data mart**: A data mart filled exclusively from the enterprise DW and its reconciled data

- **Operational data store** (**ODS**): An integrated, subject-oriented, updatable, current-valued, enterprise-wise, detailed database designed to serve operational users as they do decision support processing

Figure 11-5:
Logical data mart and @ctive data warehouse

ODS and **data warehouse** are one and the same



Near real-time ETL for
@ctive Data Warehouse

Data marts are NOT separate databases, but logical ***views*** of the DW
➔ Easier to create new data marts

2. DW Architecture: @ctive data warehouse

- **@ctive data warehouse**: An EDW that accepts *near-real-time feeds* of transactional data from the systems of record; analyzes warehouse data; and in *near-real-time relays* business rules to the DW and systems of record so that *immediate actions* can be taken in response to business events

Table 11-2: Data Warehouse vs. Data Mart

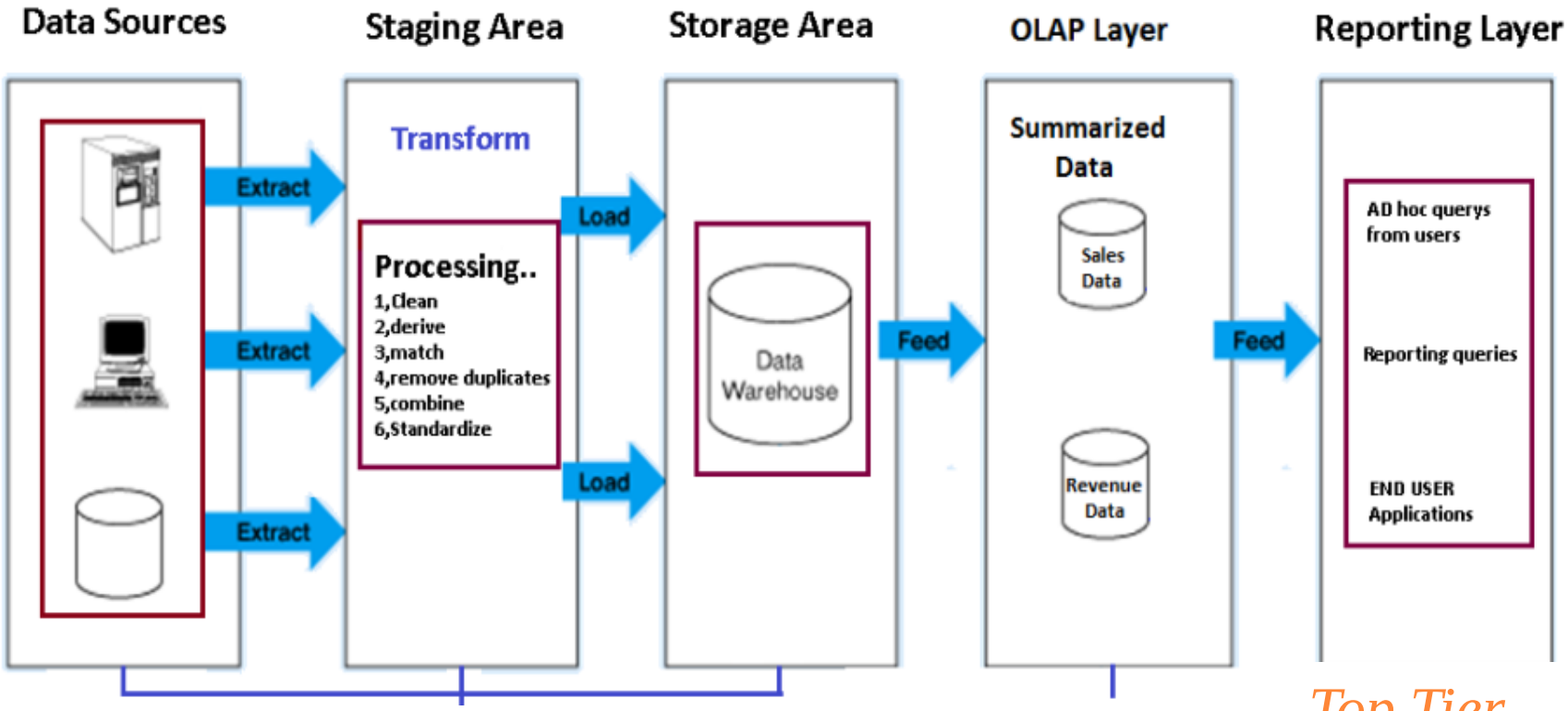
Table 11-2 Data Warehouse Versus Data Mart

Data Warehouse	*Data Mart*
Scope	*Scope*
• Application independent	• Specific DSS application
• Centralized, possibly enterprise-wide	• Decentralized by user area
• Planned	• Organic, possibly not planned
Data	*Data*
• Historical, detailed, and summarized	• Some history, detailed, and summarized
• Lightly denormalized	• Highly denormalized
Subjects	*Subjects*
• Multiple subjects	• One central subject of concern to users
Sources	*Sources*
• Many internal and external sources	• Few internal and external sources
Other Characteristics	*Other Characteristics*
• Flexible	• Restrictive
• Data-oriented	• Project-oriented
• Long life	• Short life
• Large	• Start small, becomes large
• Single complex structure	• Multi, semi-complex structures, together complex

Adapted from Strange (1997)

Source: adapted from Strange (1997).

2. DW Architecture: 3-Tier Architecture



Bottom Tier – DW layer

Middle Tier – OLAP layer

Top Tier – Front end tools

Source: DW Knowledge hub, Accessed Aug. 2015

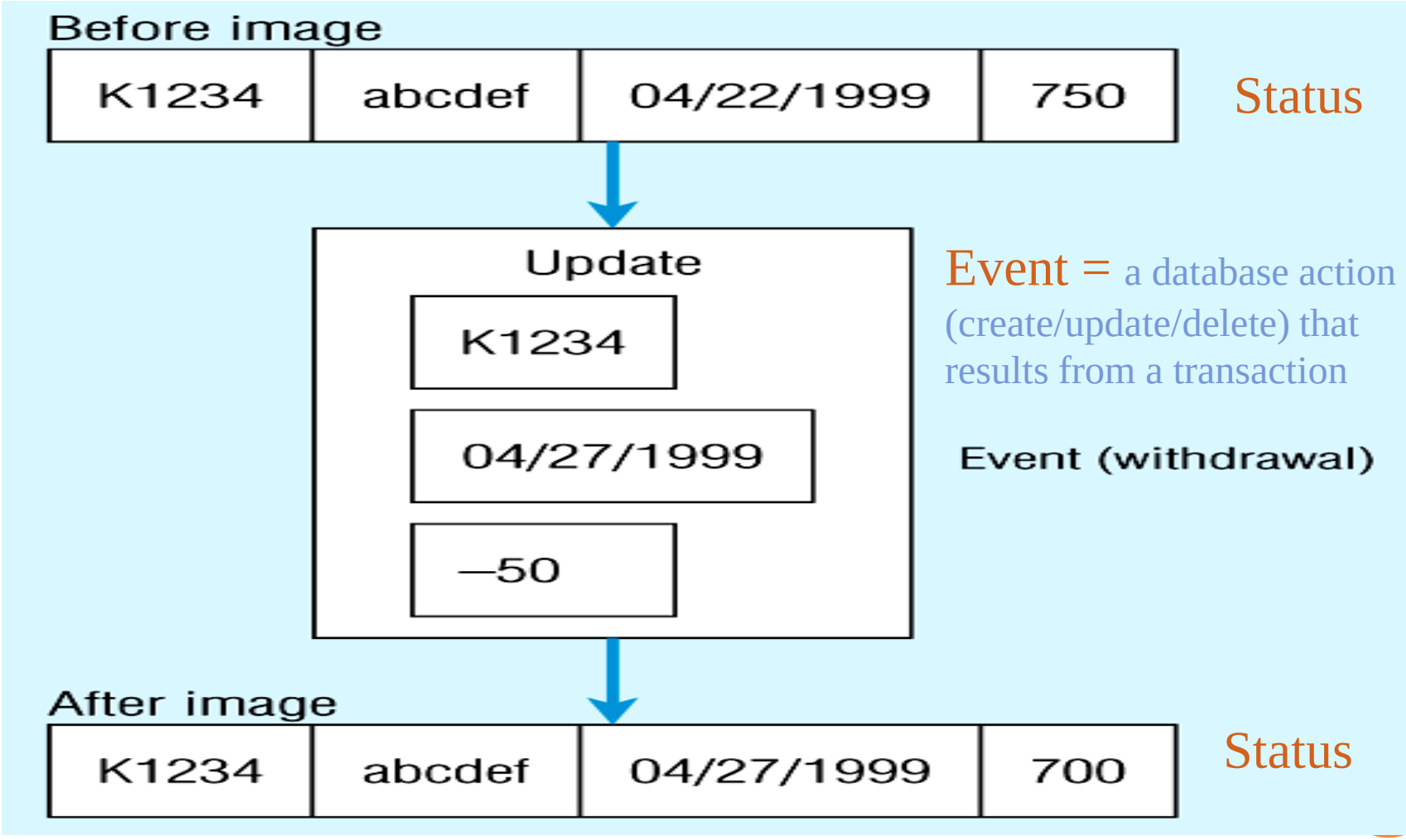
2. DW Architecture: 3-Tier Architecture

Additional Middle Tier: Middle tier is also called as application tier

- Middle is the OLAP layer that is added between bottom layer and Top tier layer
- In OLAP layer the data from DW is aggregated and stored in a multidimensional cubes
- This OLAP layer helps the reporting queries to run fast

3. Data Characteristics Status vs. Event Data

Figure 11-7: Example of DBMS log entry



3. Data Characteristics

Transient vs. Periodic Data

Figure 11-8:
Transient operational data

Changes to existing records are written over previous records, thus destroying the previous data content

Table X (10/01)

Key	A	B
001	a	b
002	c	d
003	e	f
004	g	h

Table X (10/02)

Key	A	B
001	a	b
002	r	d
003	e	f
004	y	h
005	m	n

Table X (10/03)

Key	A	B
001	a	b
002	r	d
003	e	t
005	m	n

3. Data Characteristics

Transient vs. Periodic Data

Figure 11-9:
Periodic warehouse data

Table X (10/01)

Key	Date	A	B	Action
001	10/01	a	b	C
002	10/01	c	d	C
003	10/01	e	f	C
004	10/01	g	h	C

Table X (10/02)

Key	Date	A	B	Action
001	10/01	a	b	C
002	10/01	c	d	C
002	10/02	r	d	U
003	10/01	e	f	C
004	10/01	g	h	C
004	10/02	y	h	U
005	10/02	m	n	C

Table X (10/03)

Key	Date	A	B	Action
001	10/01	a	b	C
002	10/01	c	d	C
002	10/02	r	d	U
003	10/01	e	f	C
003	10/03	e	f	U
004	10/01	g	h	C
004	10/02	y	h	U
004	10/03	y	h	D
005	10/02	m	n	C

Data is never physically altered or deleted once it has been added to the store

4. Data Reconciliation

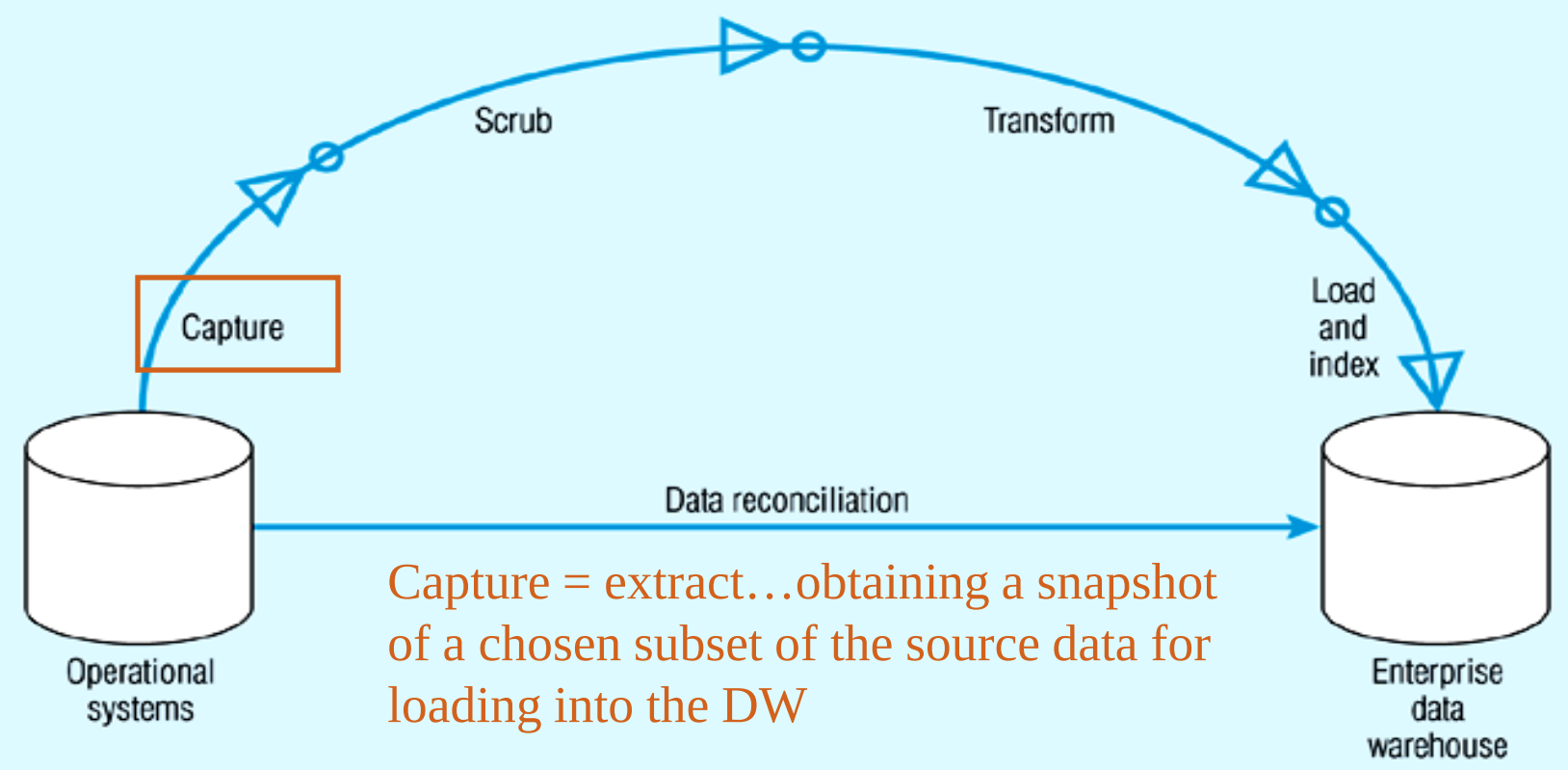
- Typical operational data is:
  - Transient – not historical
  - Not normalized (perhaps due to denormalization for performance)
  - Restricted in scope – not comprehensive
  - Sometimes poor quality – inconsistencies and errors
- After ETL, data should be:
  - Detailed – not summarized yet
  - Historical – periodic
  - Normalized – 3rd normal form or higher
  - Comprehensive – enterprise-wide perspective
  - Quality controlled – accurate with full integrity

4. The ETL Process

- Capture
- Scrub or data cleansing
- Transform
- Load and Index

ETL = Extract, transform, and load

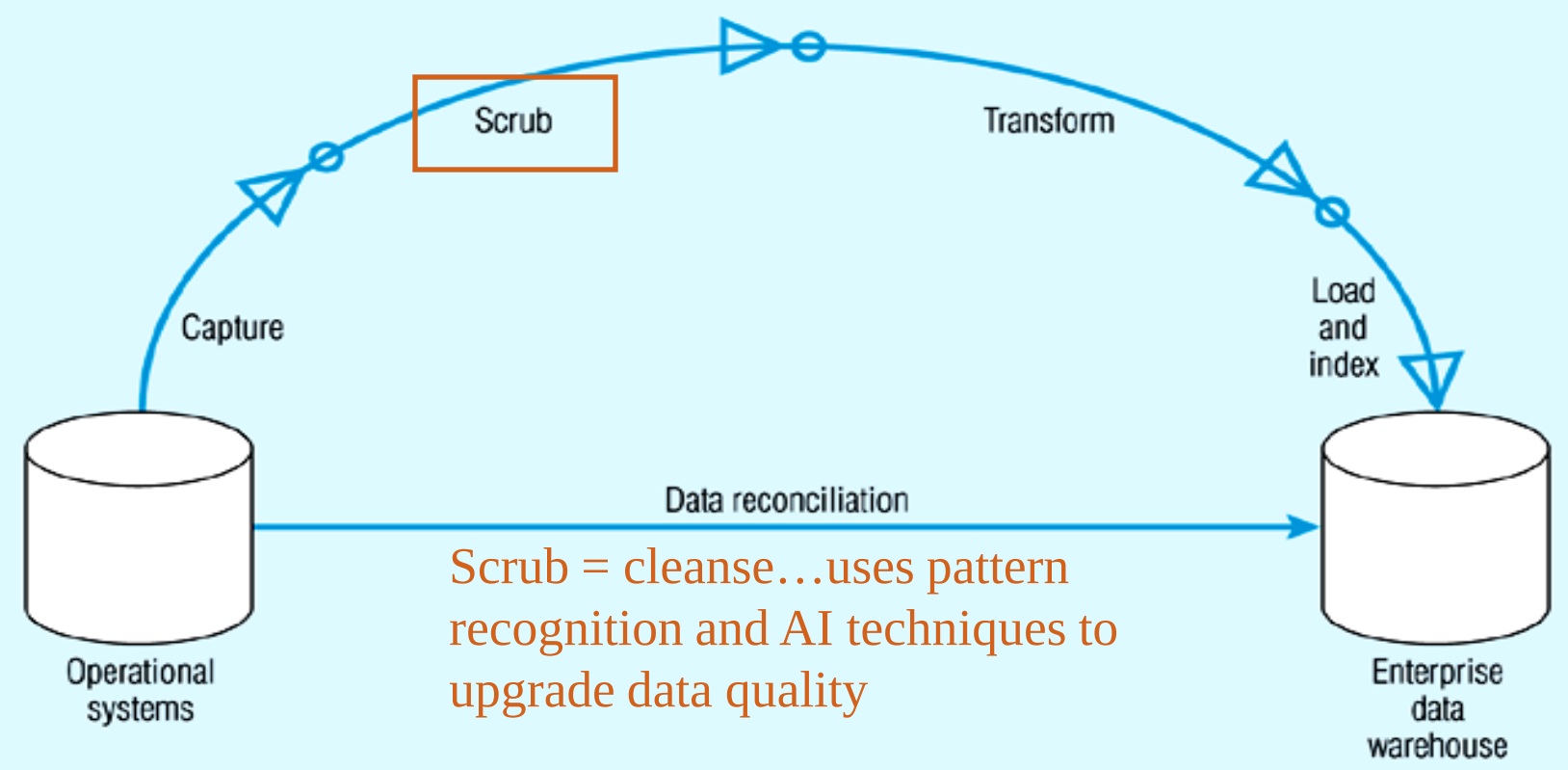
Figure 11-10: Steps in data reconciliation



Static extract = capturing a snapshot of the source data at a point in time

Incremental extract = capturing changes that have occurred since the last static extract

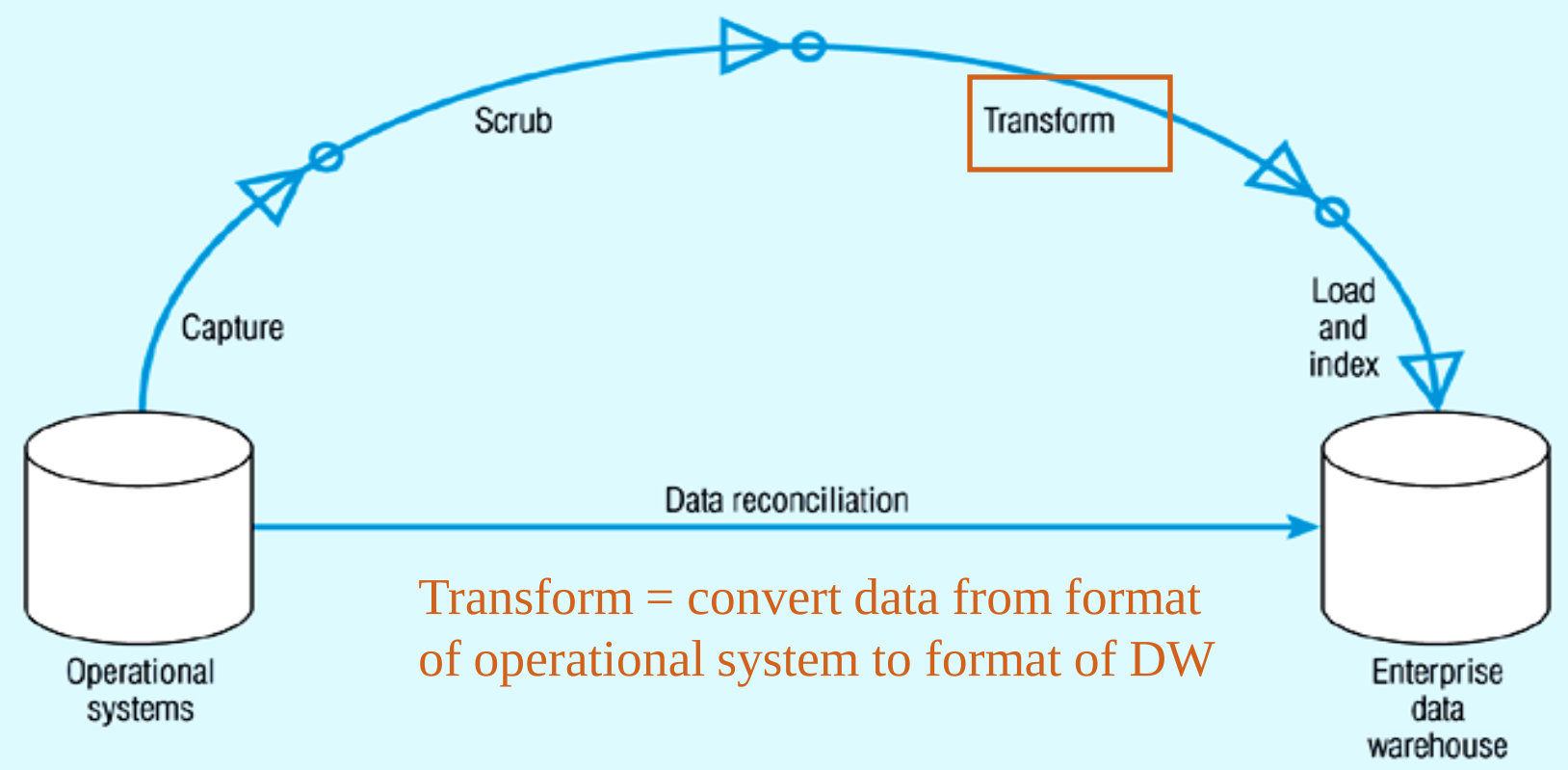
Figure 11-10: Steps in data reconciliation (continued)



Fixing errors: misspellings, erroneous dates, incorrect field usage, mismatched addresses, missing data, duplicate data, inconsistencies

Also: decoding, reformatting, time stamping, conversion, key generation, merging, error detection/logging, locating missing data

Figure 11-10: Steps in data reconciliation (continued)



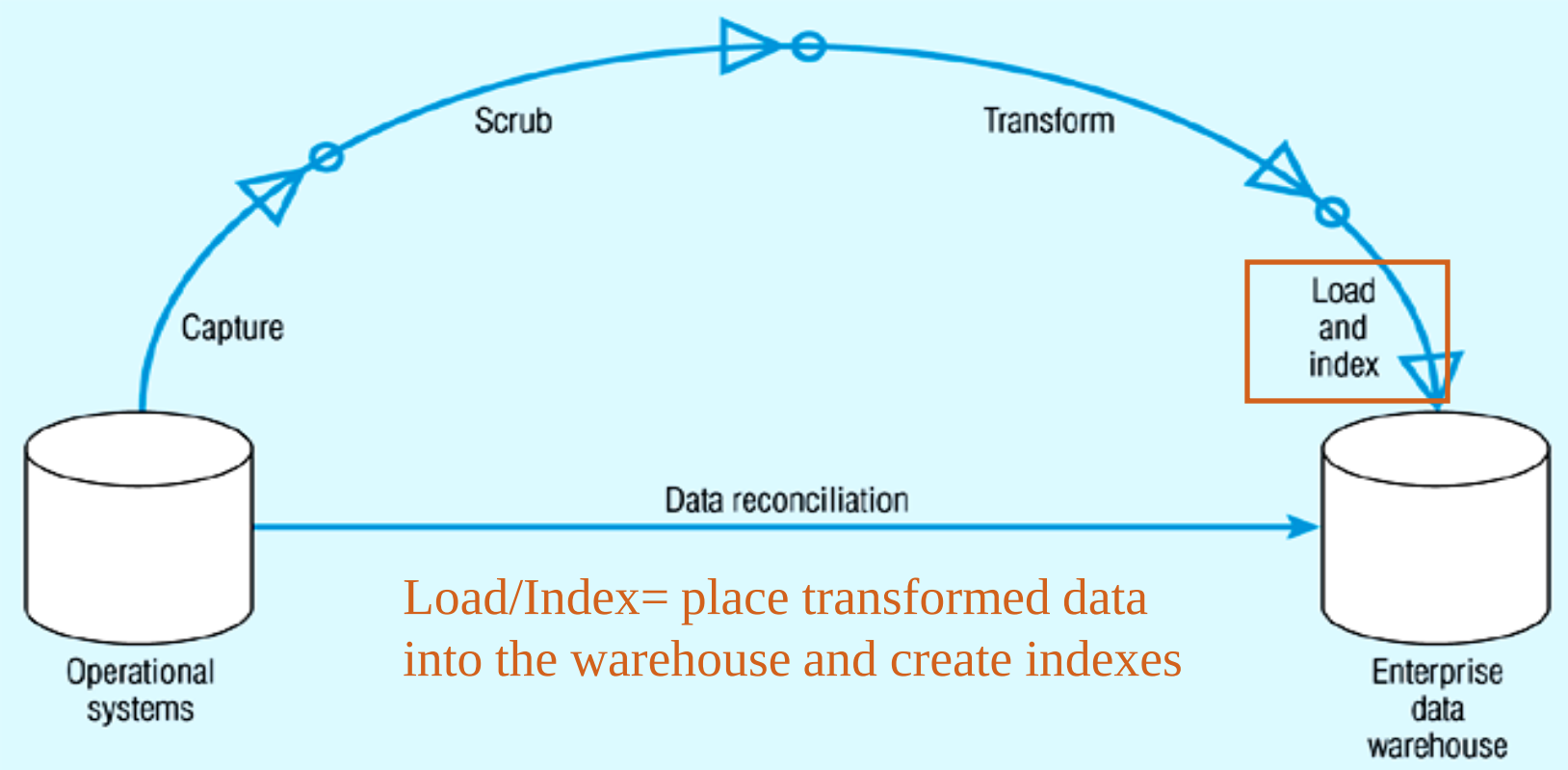
Record-level:

Selection – data partitioning
Joining – data combining
Aggregation – data summarization

Field-level:

single-field – from one field to one field
multi-field – from many fields to one, or one field to many

Figure 11-10: Steps in data reconciliation (continued)



Refresh mode: bulk rewriting of target data at periodic intervals

Update mode: only changes in source data are written to DW

5. Data Transformation

- Data transformation is the component of data reconciliation that converts data from the format of the source operational systems to the format of enterprise DW

- Consists of a variety of different functions:
  - record-level functions
  - field-level functions
  - more complex transformation.

5. Record-level functions & Field-level functions

- **Record-level functions**
  - Selection: data partitioning
  - Joining: data combining
  - Normalization
  - Aggregation: data summarization

- **Field-level functions**
  - Single-field transformation: from one field to one field
  - Multi-field transformation: from many fields to one, or one field to many

Figure 11-11: Single-field transformation

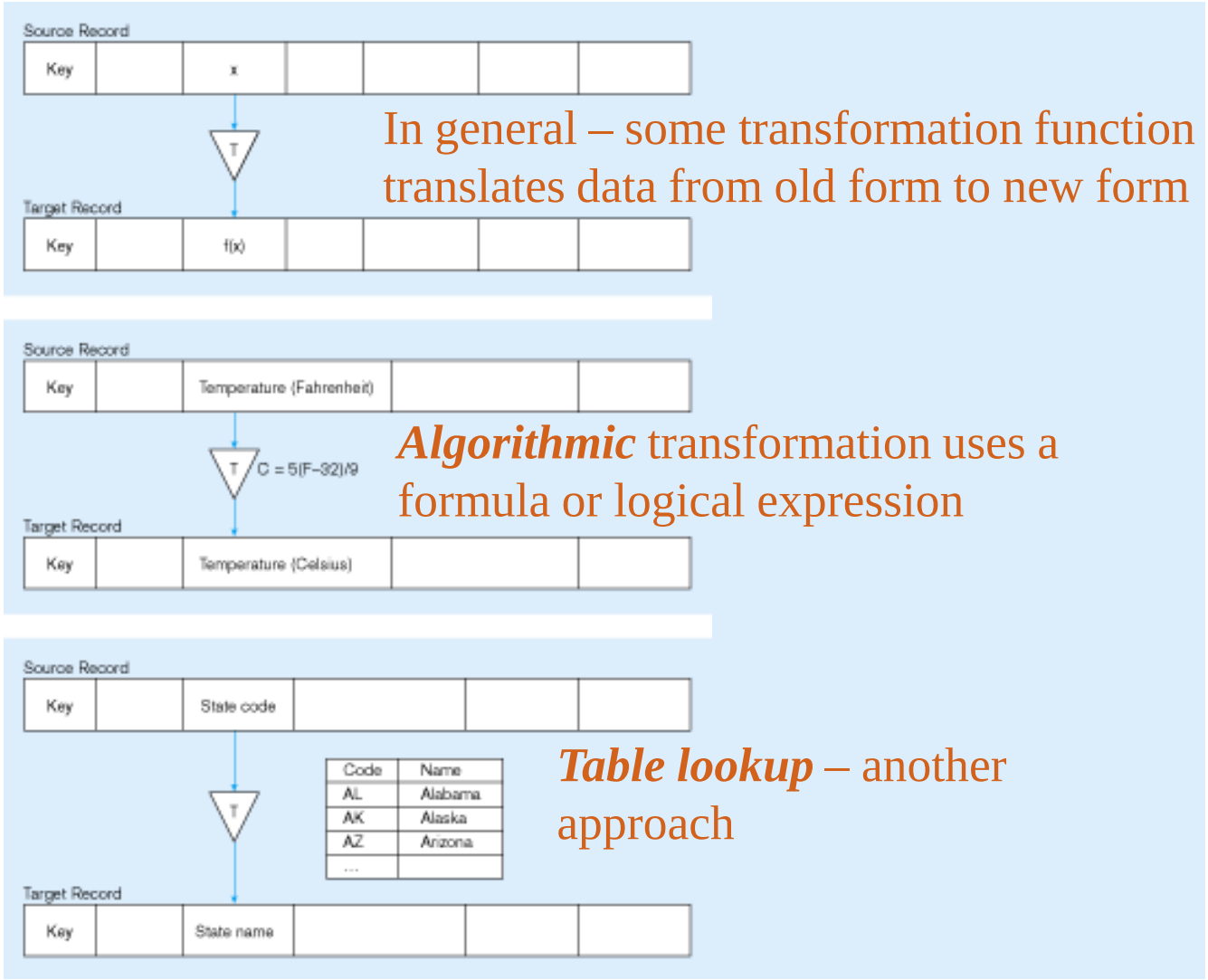
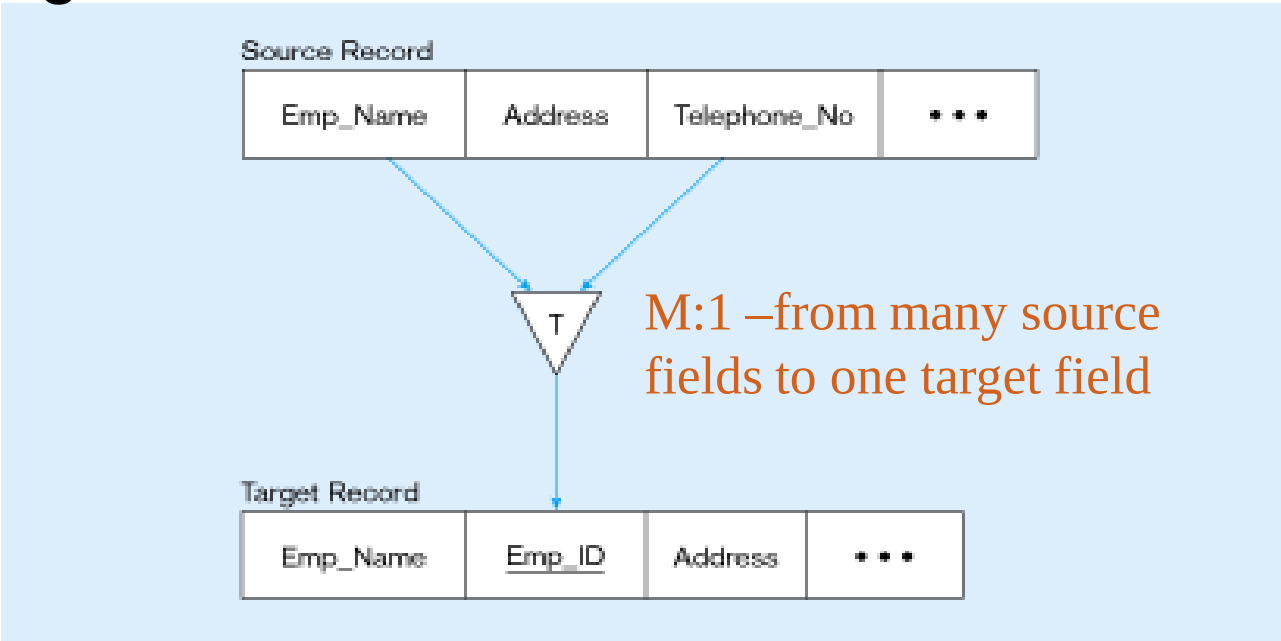
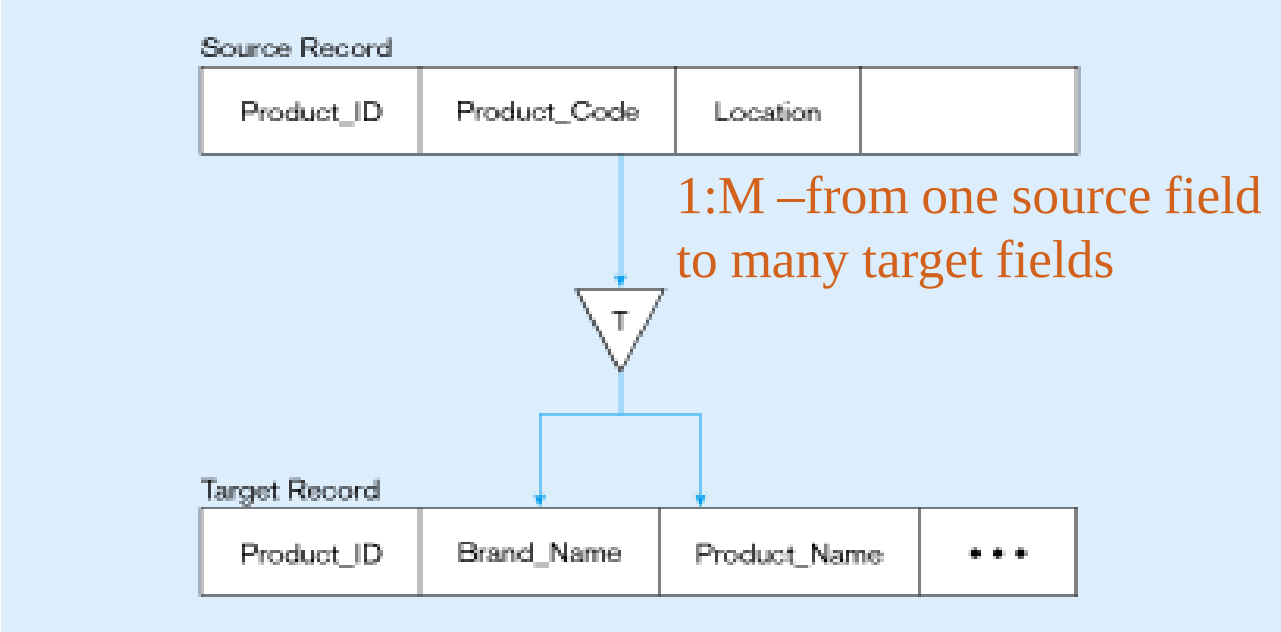


Figure 11-12: Multifield transformation





6. Derived Data

- Objectives
  - Ease of use for decision support applications
  - Fast response to predefined user queries
  - Customized data for particular target audiences
  - Ad-hoc query support
  - Data mining capabilities

- ➔ Characteristics
  - Detailed (mostly periodic) data
  - Aggregate (for summary)
  - Distributed (to departmental servers)

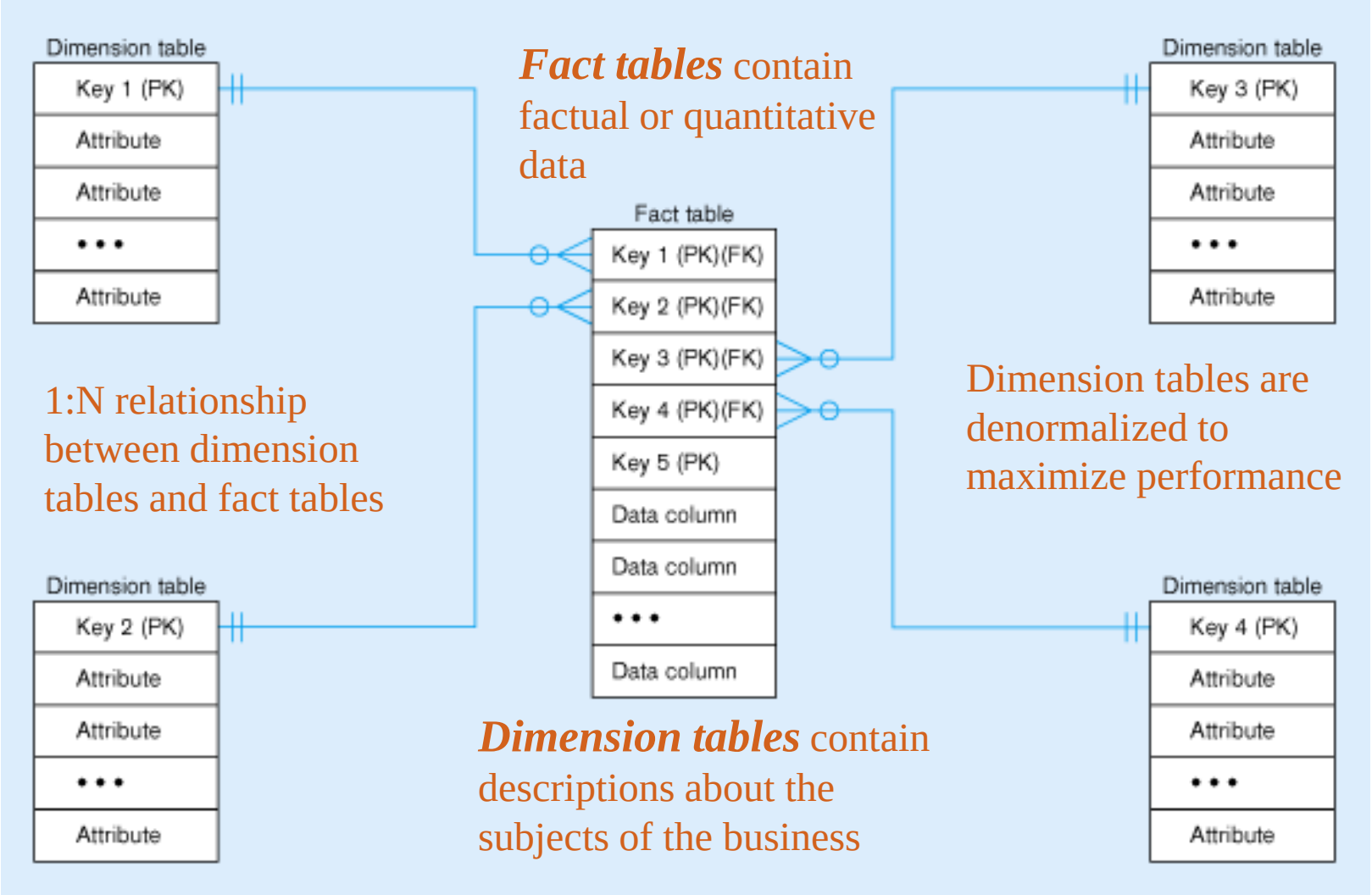
Most common data model = **star schema**
(also called "dimensional model")

6. The Star Schema

- *Star schema*: is a simple database design in which dimensional data (describing how data are commonly aggregated) are separated from fact or event data

- A star schema consists of two types of tables: ***fact tables*** and ***dimension tables***

Figure 11-13: Components of a **star schema**



Excellent for ad-hoc queries, but…
bad for online transaction processing

Figure 11-14: Star schema example

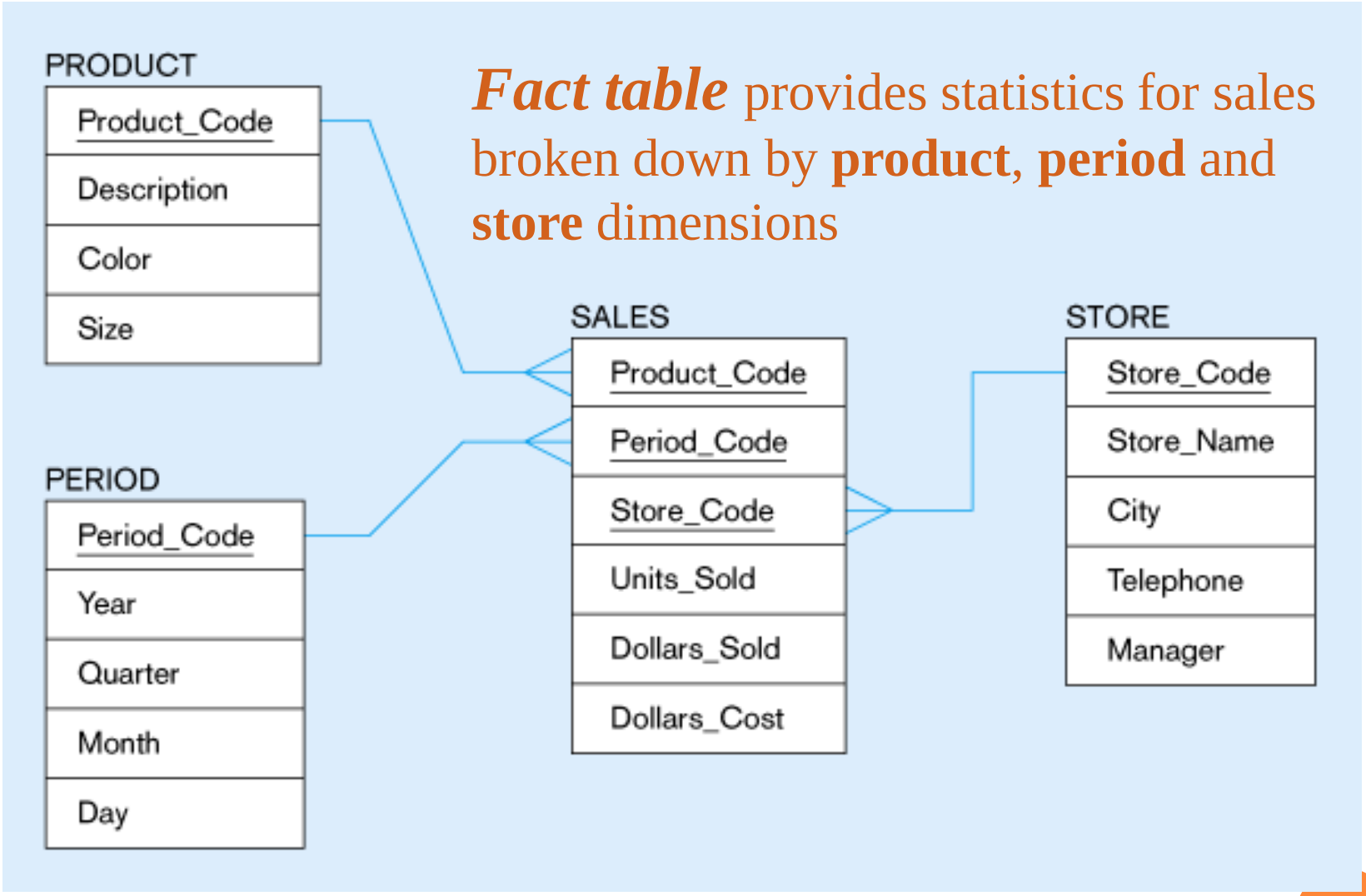
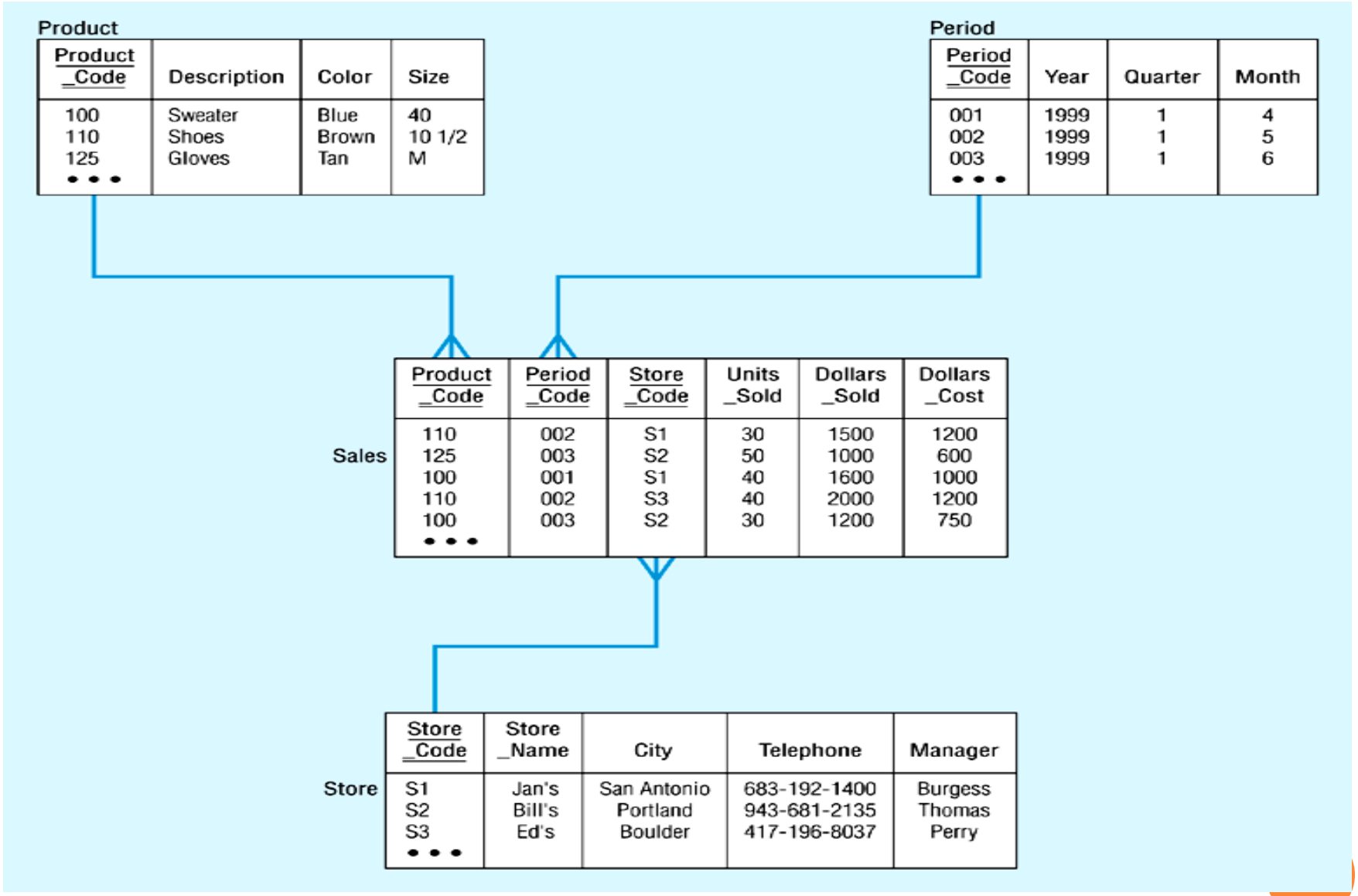


Figure 11-15: Star schema with sample data



6. Issues Regarding Star Schema

- Dimension table keys must be ***surrogate*** (non-intelligent and non-business related), because:
  - Keys may change over time
  - Length/format consistency

- Granularity of Fact Table – what level of detail do you want?
  - Transactional grain – finest level
  - Aggregated grain – more summarized
  - Finer grain ➔ better ***market basket analysis*** capability
  - Finer grain ➔ more dimension tables, more rows in fact table

6. Issues Regarding Star Schema

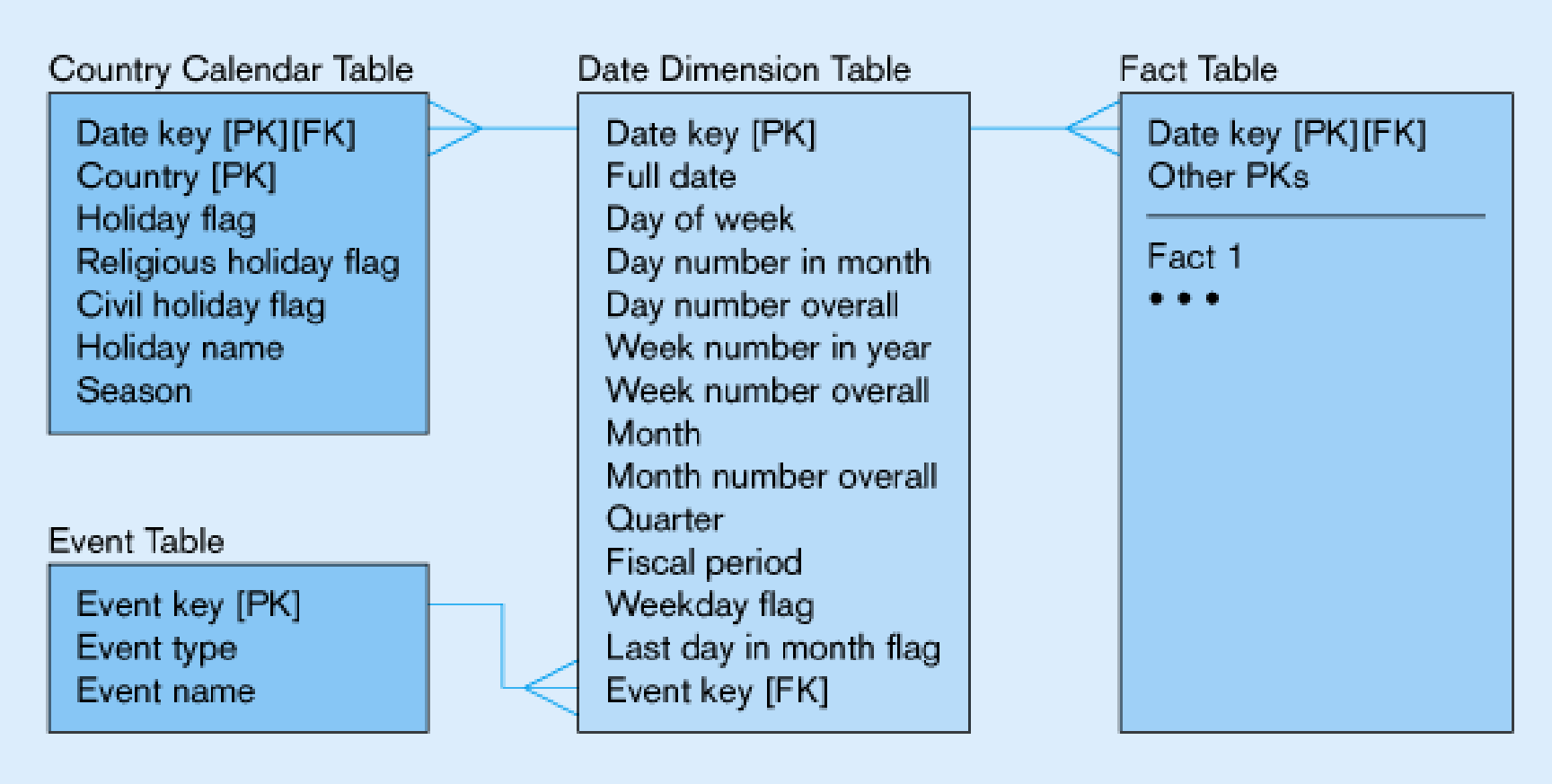
- Duration of the database
  - Ex: 13 months or 5 quarters
  - Some businesses need for a longer duration

- Size of the fact table
  - Estimate the number of possible values for each dimension associated with the fact table
  - Multiply the values obtained in the first step after making any necessary adjustments

Ex., 100 products, 4 periods (for quarters in a year), 3 stores => |Fact table| = 100*4*3 = 1,200 rows;

|Fact table for **5 years**| = 100*4***5***3 = 6,000 rows;

Figure 11-16: Modeling dates



Fact tables contain time-period data

➔ Date dimensions are important

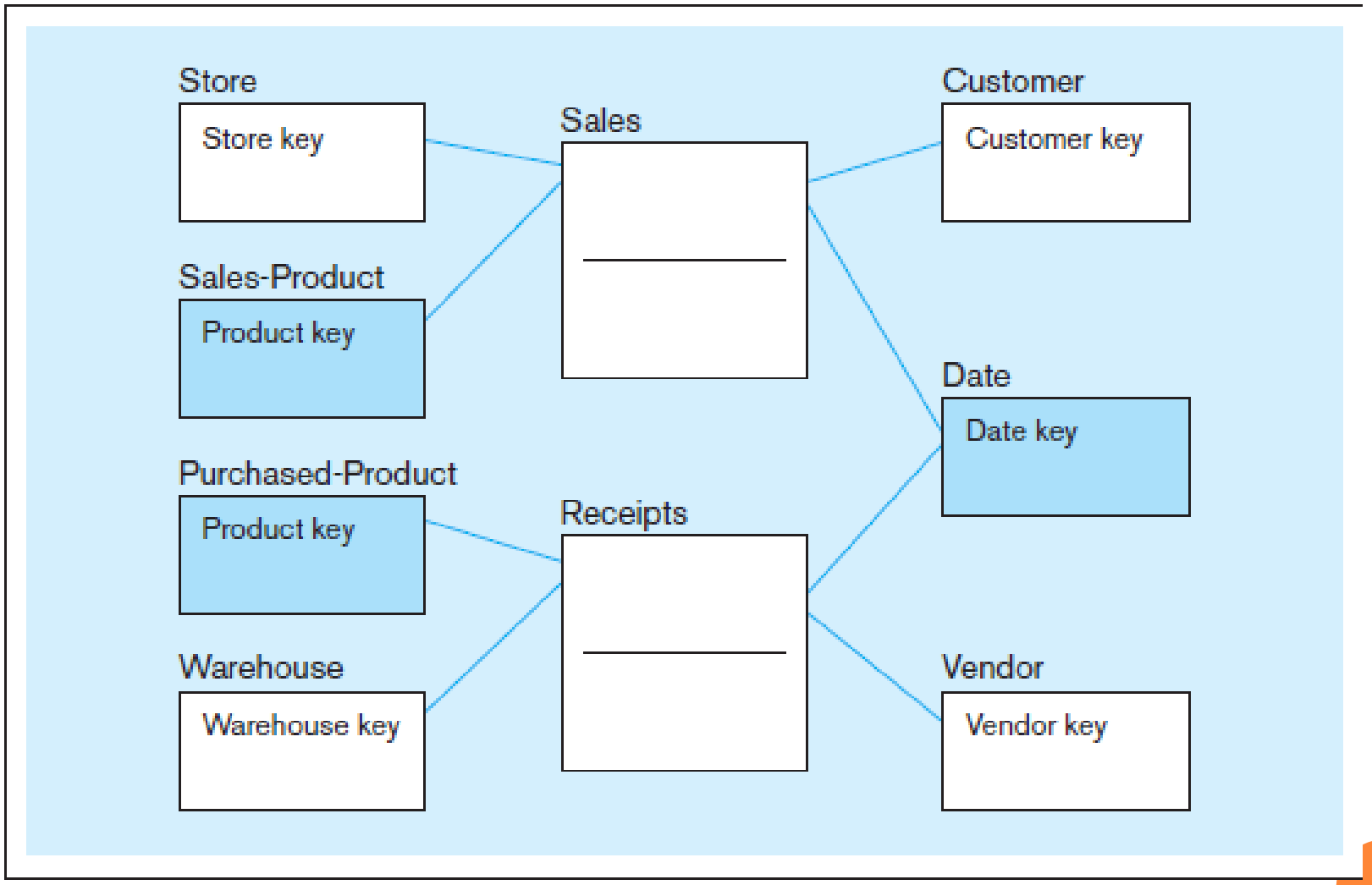
6. Variations of the Star Schema

- 1. Multiple fact tables
- 2. Factless fact tables
- 3. Normalizing Dimension Tables
- 4. Snowflake schema

6.1. Multiple Fact tables

- More than one fact table in a given star schema
- Ex: There are 2 fact tables, one at the center of each star:
  - *Sales* – facts about the sale of a **product** to a **customer** in a **store** on a **date**
  - *Receipts* - facts about the receipt of a **product** from **a vendor** to a **warehouse** on a **date**
  - Two separate *product* dimension tables have been created (**why?**)
  - One *date* dimension table is used

6.1. Multiple Fact tables

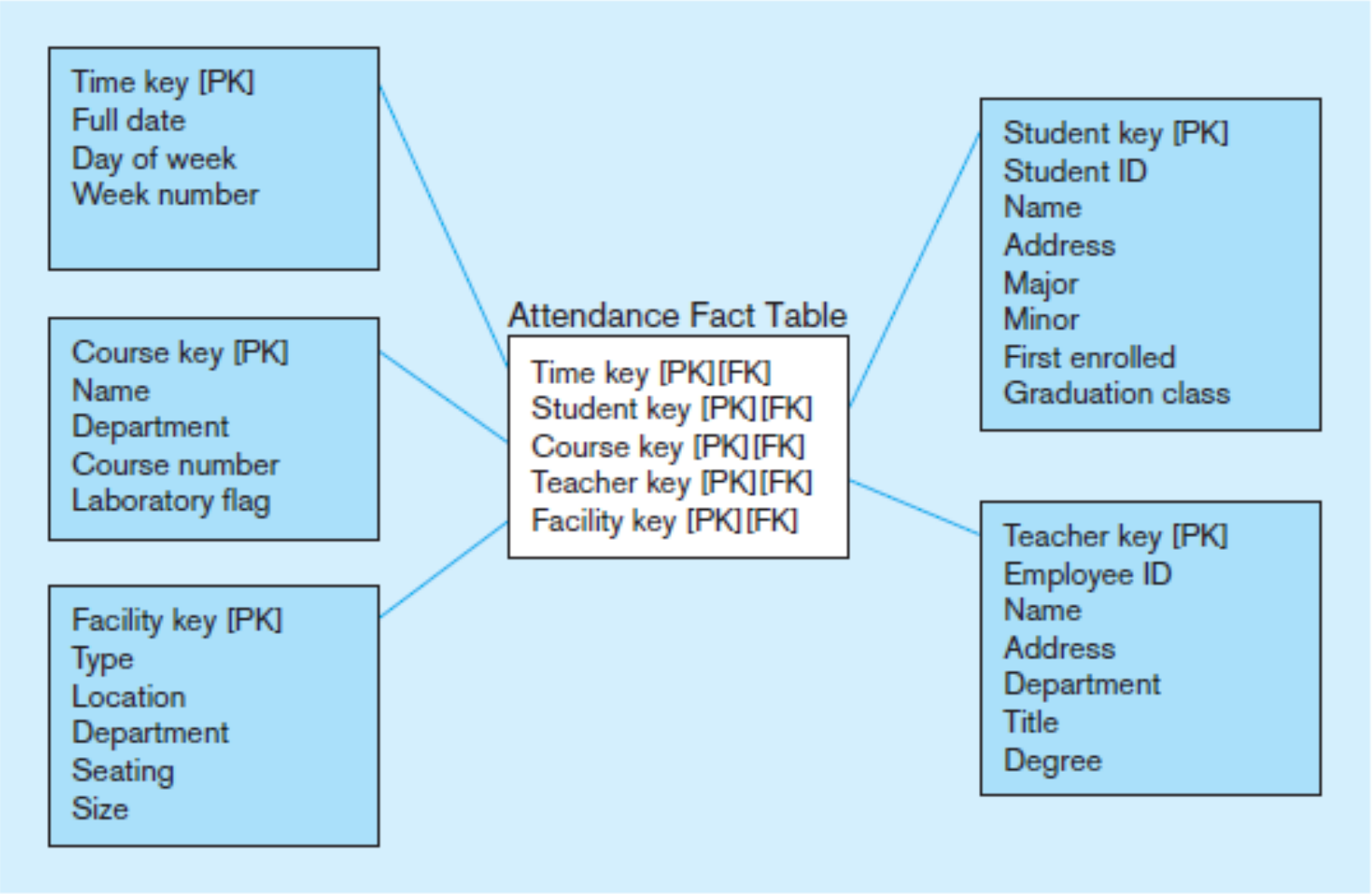


6.2. Factless Fact Tables

- There are applications in which fact tables do NOT have nonkey data but that do have foreign keys for the associated dimensions

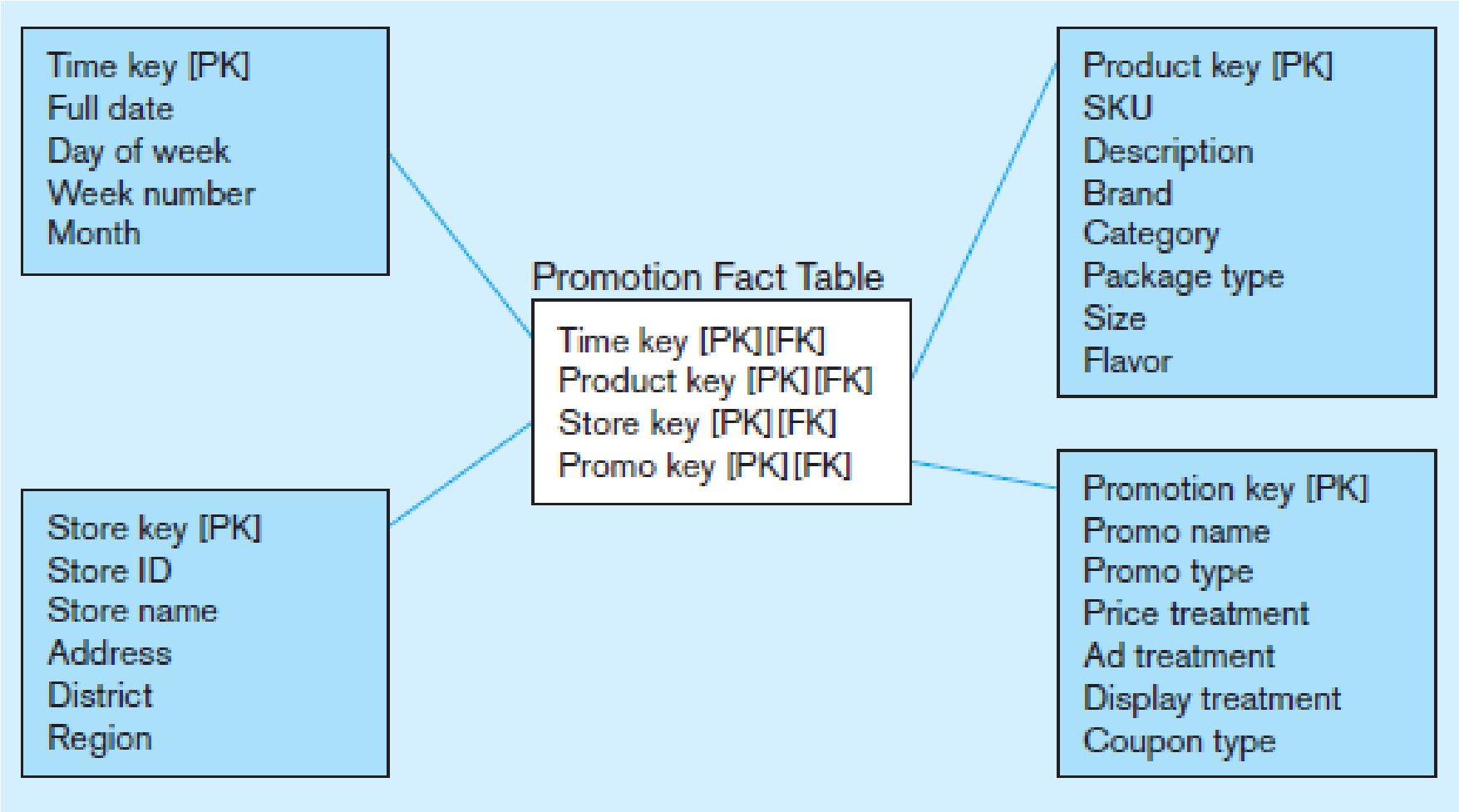
- The two situations:
  - To track events
  - To inventory the set of possible occurrences (called coverage)

6.2. Factless Fact Tables



Factless fact table showing occurrence of an event

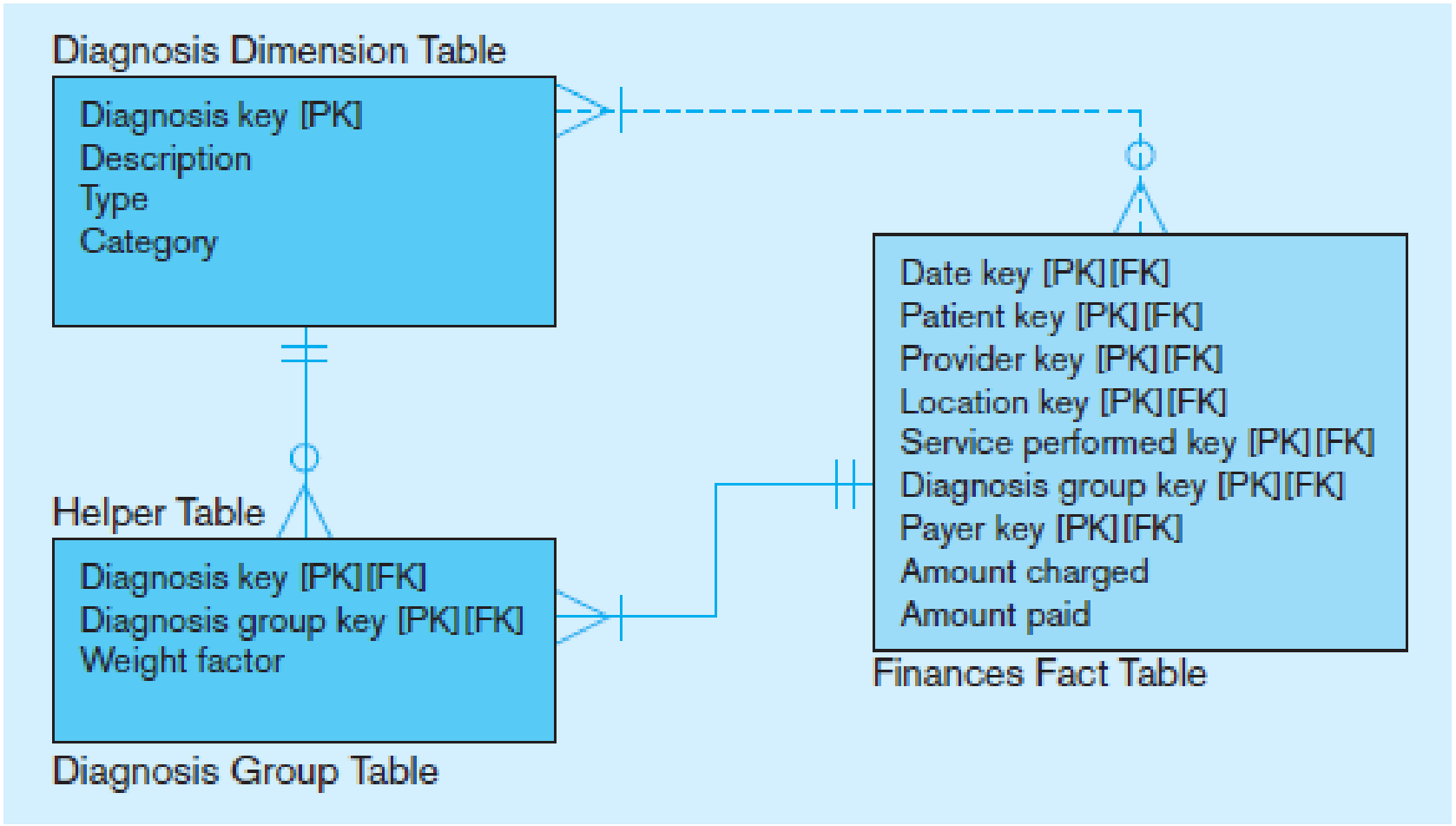
6.2. Factless Fact Tables



Factless fact table showing coverage

6.3. Normalizing dimension tables

- Dimension tables may not be normalized. Most DW experts find this acceptable
- In some situations in which it makes sense to further normalize dimension tables
- *Multivalued dimensions*:
  - Ex: Hospital charge/payment for a *patient* on a *date* is associated with one or more *diagnosis*
  - N:M relationship between the Diagnosis and Finances fact table
  - Solution: create an associative entity (*helper table*) between Diagnosis and Finances



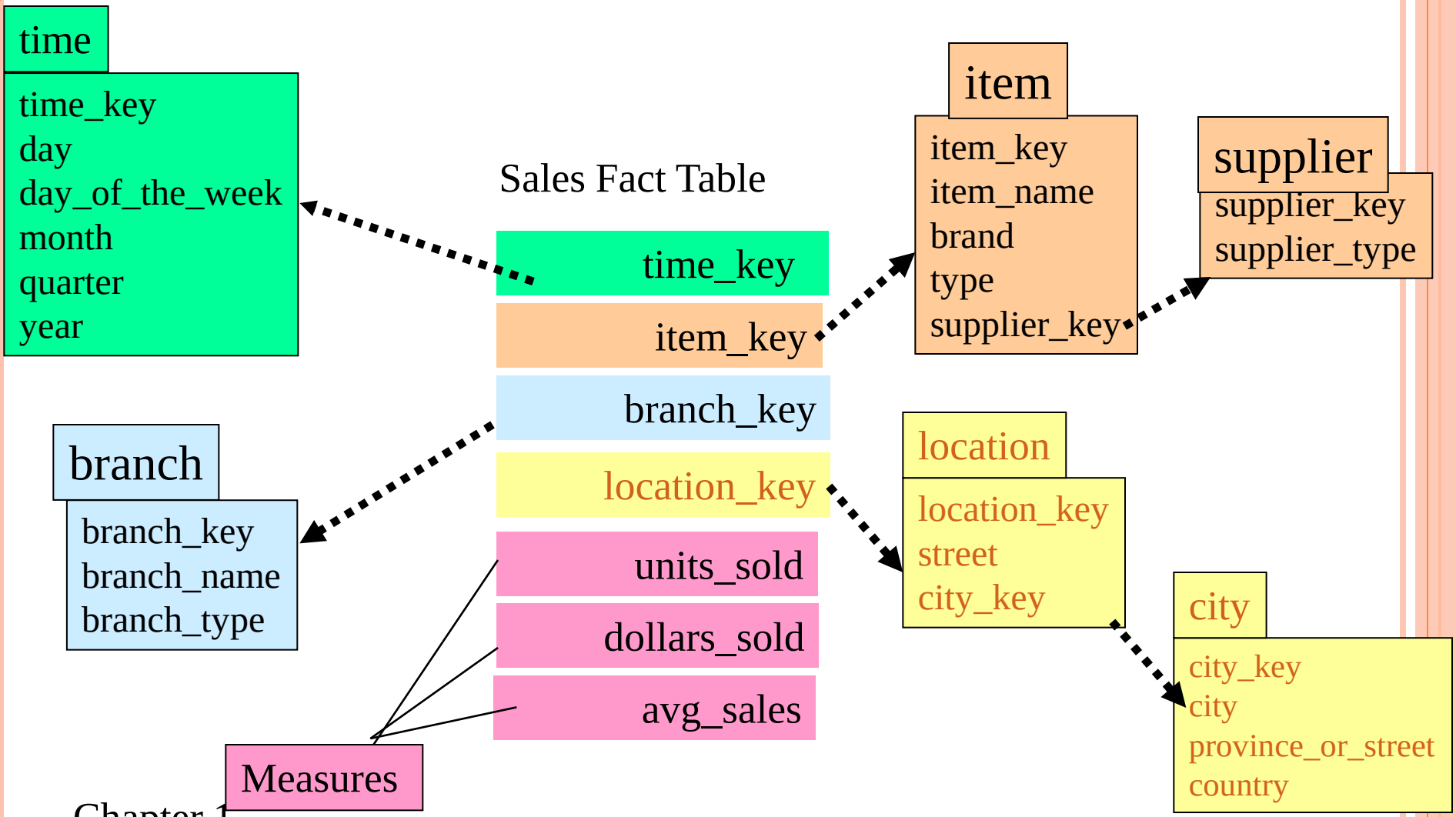
Multivalued dimension

6.4. Snowflake schema

- *Snowflake schema* is an expanded version of a star schema in which dimension tables are normalized into several related tables

- Advantages
  - Small saving in storage space
  - Normalized structures are easier to update and maintain
- Disadvantages
  - Schema less intuitive
  - Difficult to browse through the content
  - Degraded query performance because of additional joins

6.4. Example of snowflake schema



7. The User Interface

- A variety of tools are available to query and analyze data stored in DWs
  - Querying tools
  - On-line Analytical processing (OLAP, MOLAP, ROLAP) tools
  - Data Mining tools
  - Data Visualization tools

7. Role of Metadata (data catalog)

- Identify subjects of the DW/data mart
- Identify dimensions and facts
- Indicate how data is derived from enterprise DW, including derivation rules
- Indicate how data is derived from operational data store, including derivation rules
- Identify available reports and predefined queries
- Identify data analysis techniques (e.g. drill-down)
- Identify responsible people

7.1. Querying Tools

- SQL is not an analytical language
- SQL-99 includes some DW extensions, but still is not a full-featured DW querying and analysis tool
- Different DBMS vendors will implement some or all of the SQL-99 OLAP extension commands and possibly others

7.2. On-Line Analytical Processing (OLAP)

- OLAP: the use of a set of graphical tools providing users with multidimensional views of their data and allows them to analyze the data using simple windowing techniques

- ***Relational OLAP (ROLAP)***
  - OLAP tools that view the database as a traditional relational database in either a star schema or other normalized or denormalized set of tables

- ***Multidimensional OLAP (MOLAP)***
  - OLAP tools that load data into an intermediate structure, usually a three or higher dimensional array (Cube structure)

7.2. From tables to data cubes

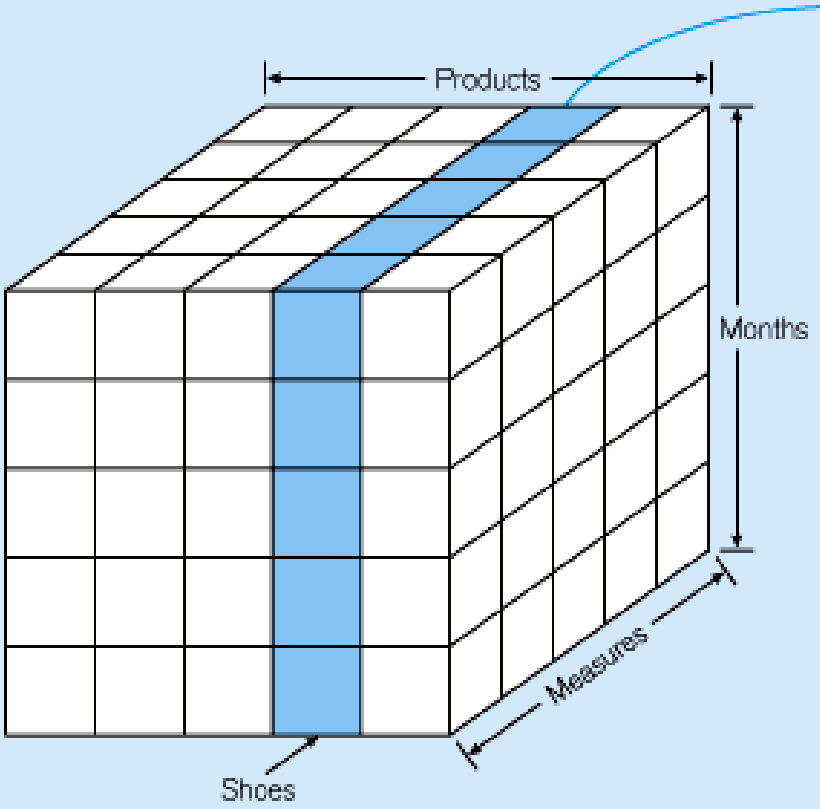
- A DW is based on a multidimensional data model which views data in the form of a data cube

- A data cube, such as *sales*, allows data to be modeled and viewed in multiple dimensions
  - Dimension tables, such as *item* (*item_name, brand, type*), or *time* (*day, week, month, quarter, year*)
  - Fact table contains measures (such as *dollars_sold*) and keys to each of the related dimension tables

7.2. MOLAP Operations

- Roll up (drill-up): summarize data
  - *by climbing up hierarchy or by dimension reduction*
- Drill down (roll down): reverse of roll-up
  - *from higher level summary to lower level summary or detailed data, or introducing new dimensions*
- Slice and dice: *project and select*

Figure 11-22: Slicing a data cube



	Measure		
	Units	Revenue	Cost
January	250	1564	1020
February	200	1275	875
March	350	1800	1275
April	400	1935	1500
May	485	2000	1560

Product: Shoes

Figure 11-23: Example of drill-down

Summary report

Brand	Package size	Sales
SofTowel	2-pack	$75
SofTowel	3-pack	$100
SofTowel	6-pack	$50

Drill-down with color added

Brand	Package size	Color	Sales
SofTowel	2-pack	White	$30
SofTowel	2-pack	Yellow	$25
SofTowel	2-pack	Pink	$20
SofTowel	3-pack	White	$50
SofTowel	3-pack	Green	$25
SofTowel	3-pack	Yellow	$25
SofTowel	6-pack	White	$30
SofTowel	6-pack	Yellow	$20

7.2. OLAP tool Vendors

- IBM
- Informix
- Cartelon
- NCR
- Oracle (Oracle Warehouse builder, Oracle OLAP)
- Red Brick
- Sybase
- SAS
- Microsoft (SQL Server OLAP)
- Microstrategy Corporation

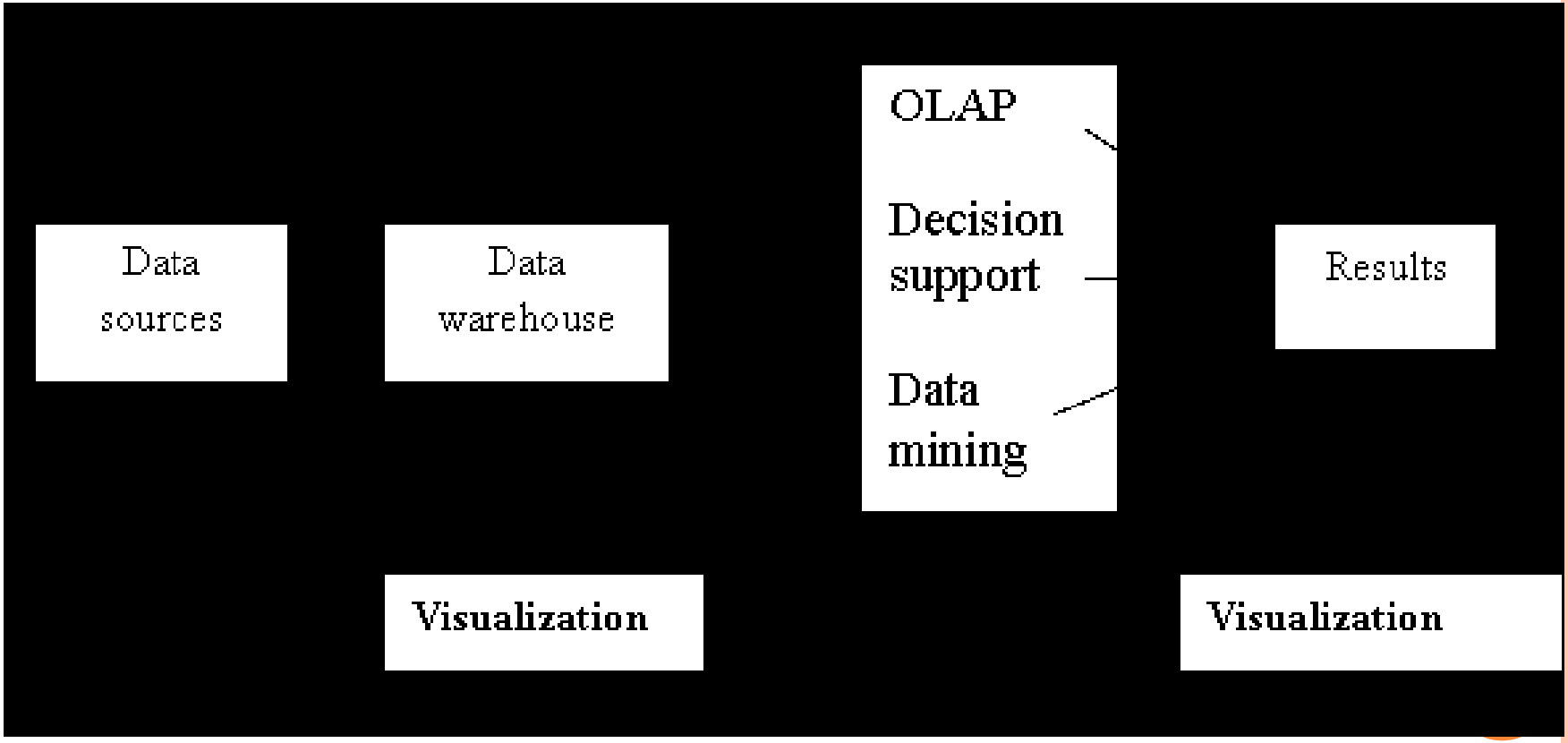
7.3. Data Mining

- Is knowledge discovery using a blend of statistical, AI, and computer graphics techniques

- Goals:
  - Explain observed events or conditions
  - Confirm hypotheses
  - Explore data for new or unexpected relationships
- Techniques
  - Case-based reasoning
  - Rule discovery
  - Neural networks
  - Signal processing
  - Fractals

7.4. Data Visualization

- Data visualization is the representation of data in graphical/multimedia formats for human analysis



8. Summary

- DW: A subject-oriented, integrated, time-variant, non-updatable collection of data used in support of management decision-making processes
- Data Warehousing: The process of constructing and using a DW
- DW vs. Data mart
- DW Architectures: Generic Two-Level Architecture, Independent Data Mart, Dependent Data Mart and Operational Data Store, Logical Data Mart and @ctive Warehouse, and Three-Layer Architecture

8. Summary

- Data reconciliation: Process to bring data from operational systems to the DW using the ETL process
- ETL (Extraction, Transformation, Load): *Capture, Scrub or data cleansing, Transform, Load and Index*
- Data transformation: converts data from the format of the source operational systems to the format of EDW: *record-level functions, field-level functions, more complex transformation*

8. Summary

- **Star schema:** fact table(s) and dimension tables (N-1)
- Date dimensions are important
- Star schema variation: *Multiple fact tables, Factless fact tables, Normalizing Dimension Tables, Snowflake schema*
- User interface for DW: *Querying tools, On-line Analytical processing (OLAP, MOLAP, ROLAP) tools, Data Mining tools, Data Visualization tools*